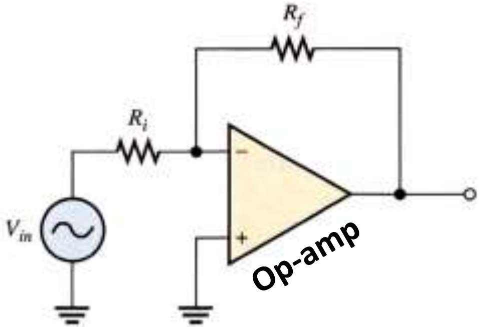
The Operational Amplifier (op-amp)



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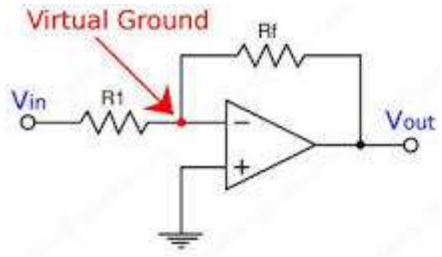
What is an Op-Amp

- Low cost integrating circuit consisting of:
  - Transistors
  - Resistors
  - Capacitors
- Able to amplify a signal due to an external power supply
- Name derives from its use to perform operations on a signal.

Virtual ground

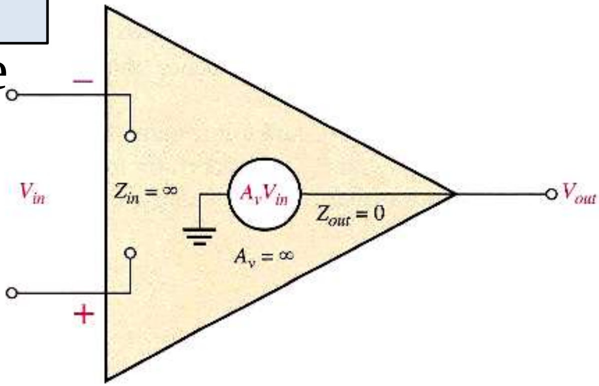
In opamps the term virtual ground means that **the voltage at that particular node is almost equal to ground voltage (0V)**. It is not physically connected to ground. This concept is very useful in analysis of opamp circuits



Characteristics of op-amp

Ideal op-amp

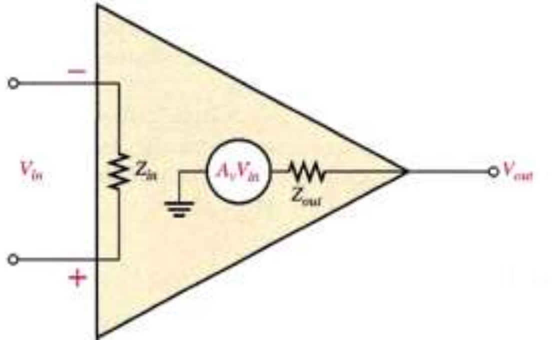
- ❑ Infinite voltage gain
- ❑ Infinite bandwidth
- ❑ Infinite input impedance
- ❑ Zero output impedance



Characteristics of op-amp

Practical op-amp

- Very high voltage gain
- Wide bandwidth
- Very high input impedance
- Very low output impedance



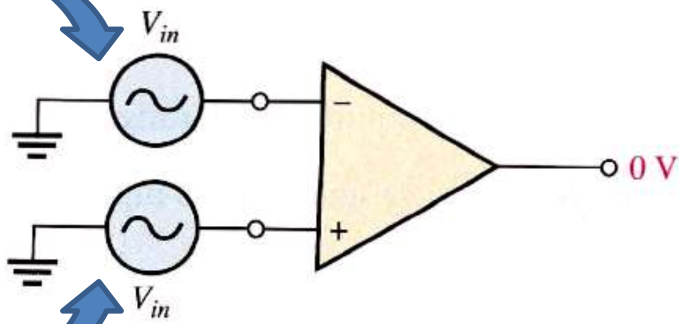
Op-amp parameters

1. Common mode input voltage
2. CMRR
3. Common mode input voltage range
4. Slew rate

Common mode input voltage

It is the voltage that appears on the both input with same

Amplitude (A)
Frequency (f)
Phase (φ)



When **equal input** signal are applied **to both inputs**, they cancel, resulting in a **zero** output voltage.

Common Mode Rejection Ratio (CMRR)

It is a measure of ability of an amplifier to reject common mode signals

Practical op-amp exhibits a very small common mode gain **(<1).**

$$\text{CMRR} = \frac{A_{ol}}{A_{cm}}$$

$$\text{CMRR} = 20 \log\left(\frac{A_{ol}}{A_{cm}}\right)$$

It measure how much higher Open-loop gain with respect to common mode gain.

Open-Loop Voltage Gain

Ratio of output voltage to input voltage when there is no external component. It is the internal voltage gain. It is set entirely by the internal design.

What does it mean by "CMRR is 100,000"?

We know that: $$\text{CMRR} = \frac{A_{ol}}{A_{cm}}$$

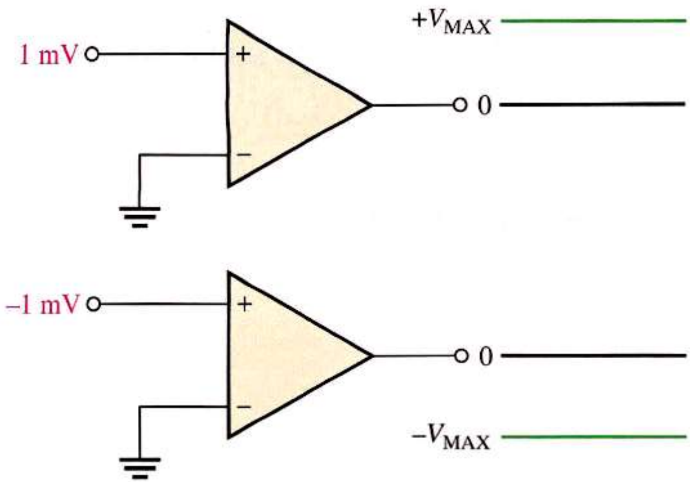
01. It means that the desired input signal (differential) is amplified 100,000 times more than the unwanted noise (common-mode)

02. If the amplitudes of the differential input signal and the common-mode noise are equal, the desired signal, the desired signal will be appeared on the output 100,000 times larger than common-mode noise.

Thus the noise/interference has been essentially eliminated.

Common mode input voltage range

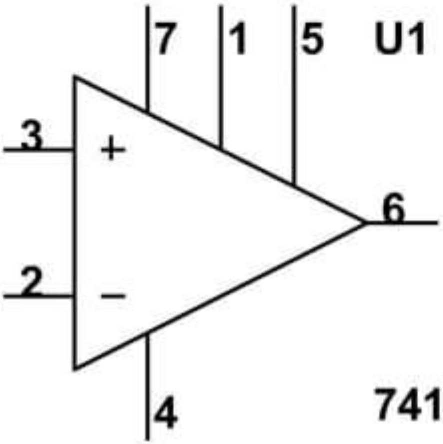
All op-amps have limitations on the range of voltages over which they will operate. The *common-mode input voltage range* is the range of input voltages which, when applied to both inputs, will not cause clipping or other output distortion. Many op-amps have common-mode input voltage ranges of ±10 V with dc supply voltages of ±15 V.



Applications of Op-Amps

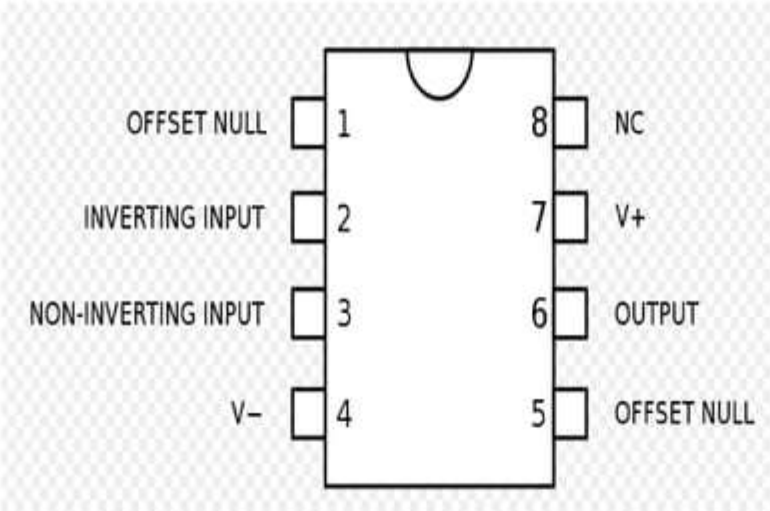
- Simple Amplifiers
- Summers
- Comparators
- Integrators
- Differentiators
- Analog to Digital Converters

Circuit Symbol and Pin Identification

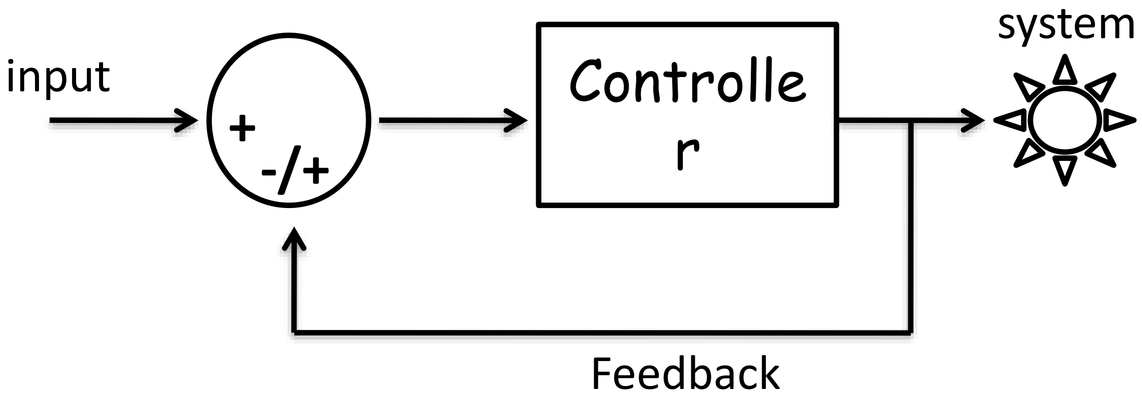


- 2 Inverting Input
- 3 Non-Inverting Input
- 6 Output
- 7 + Voltage Supply V_{CC}
- 4 – Voltage Supply V_{EE}
- 1 and 5 -- Offset Null

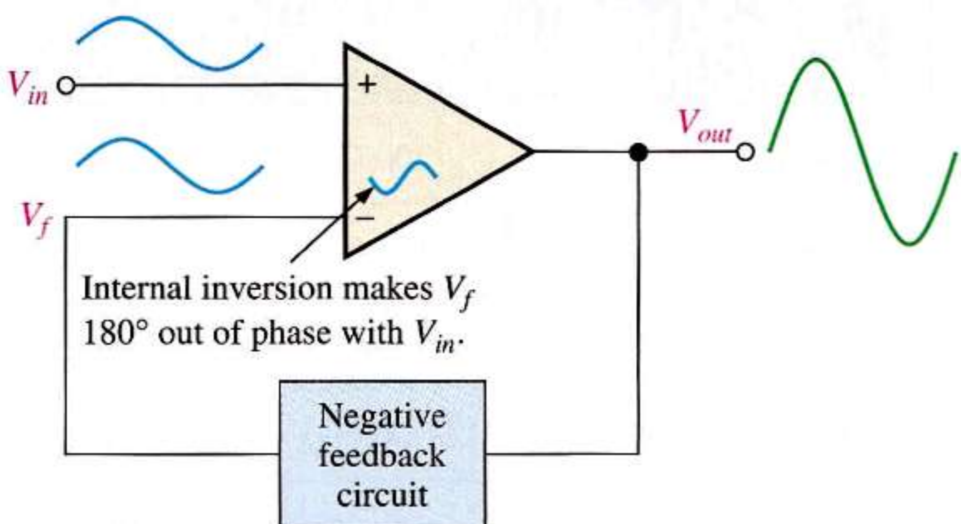
What do they really look like?

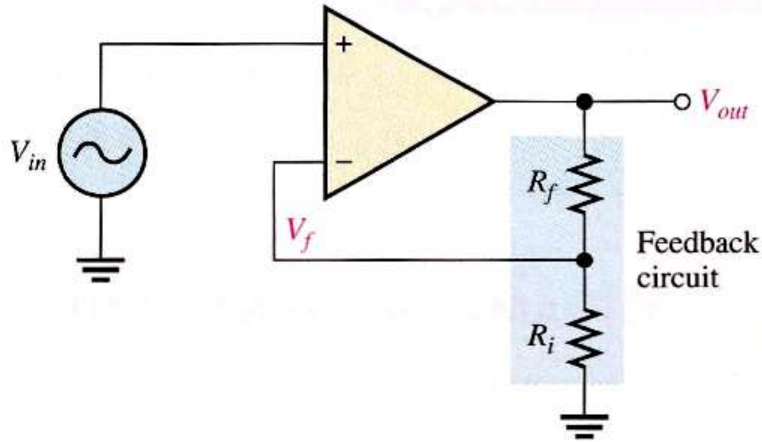


Negative Feedback



What is negative feedback?





V_{in}
V_f
V_{out}
R_f
R_i
Feedback
circuit

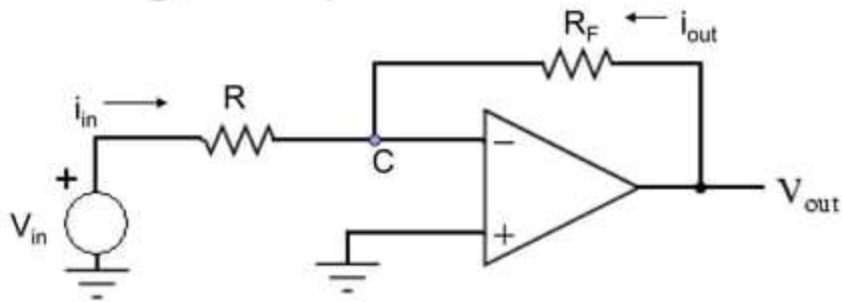
Why use negative feedback?

- Gain is reduced and can be controlled so that the op-amp can function as a linear amplifier

Non-inverting amplifier
Voltage follower
Inverting amplifier

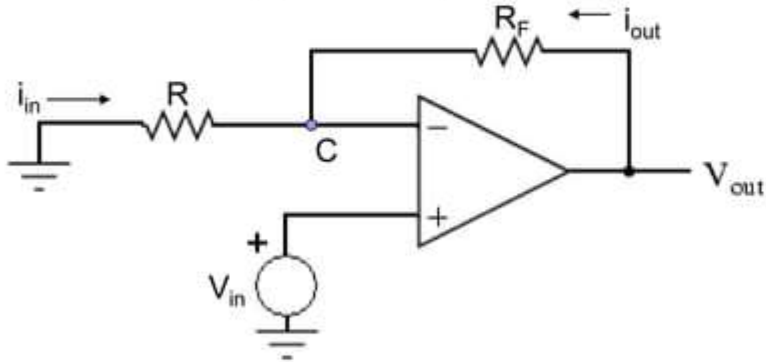
- Control of input & output impedances amplifier bandwidth

Inverting Amplifier



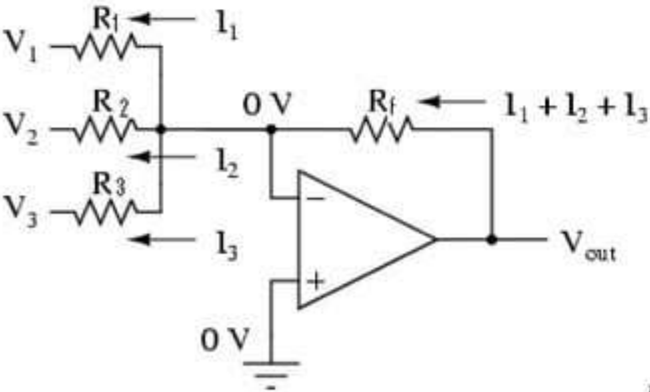
$$\frac{V_{out}}{V_{in}} = -\frac{R_F}{R}$$

Non-Inverting Amplifier



$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_F}{R}$$

Summing Circuits



- Used to add analog signals
- Voltage averaging function into summing function

Calculate closed loop gain for each input

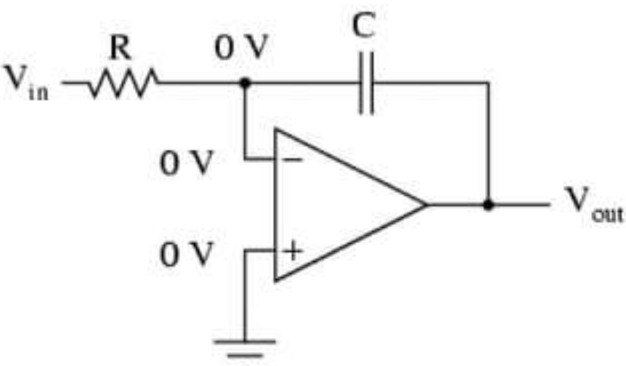
$$A_{CL1} = \frac{-R_f}{R_1} \quad A_{CL1} = \frac{-R_f}{R_2} \quad A_{CL1} = \frac{-R_f}{R_3}$$

$$V_o = V_{in} \cdot A_{CLn} \qquad V_o = -V_1 \cdot \frac{R_f}{R_1} - V_2 \cdot \frac{R_f}{R_2} - V_3 \cdot \frac{R_f}{R_3}$$

If all resistors are equal in value: $$V_o = -(V_1 + V_2 + V_3)$$

Integrating Circuit

Integrator



- Replace feedback resistor of inverting op-amp with capacitor
- A constant input signal generates a certain rate of change in output voltage
- Smoothes signals over time

$$\frac{dv_{out}}{dt} = -\frac{V_{in}}{RC}$$

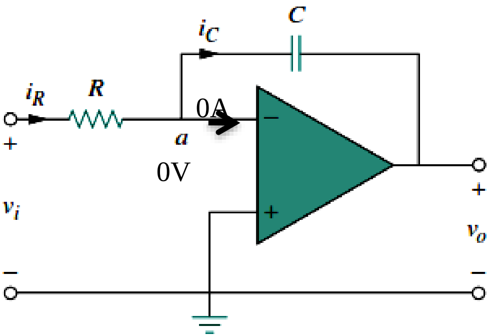
or

$$V_{out} = \int_0^t -\frac{V_{in}}{RC}\, dt + c$$

Where,

c = Output voltage at start time (t=0)

Integrator



$$i_R = \frac{v_i}{R},$$ ← Vi – 0 = Vi

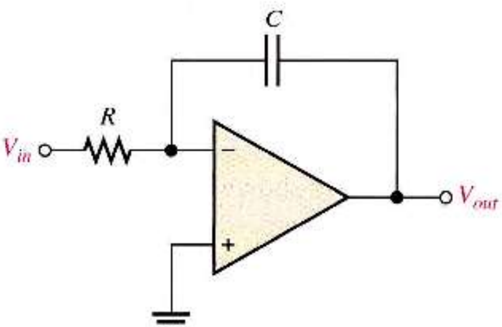
$$i_C = -C\frac{dv_o}{dt}$$ ← 0 - Vo = - Vo

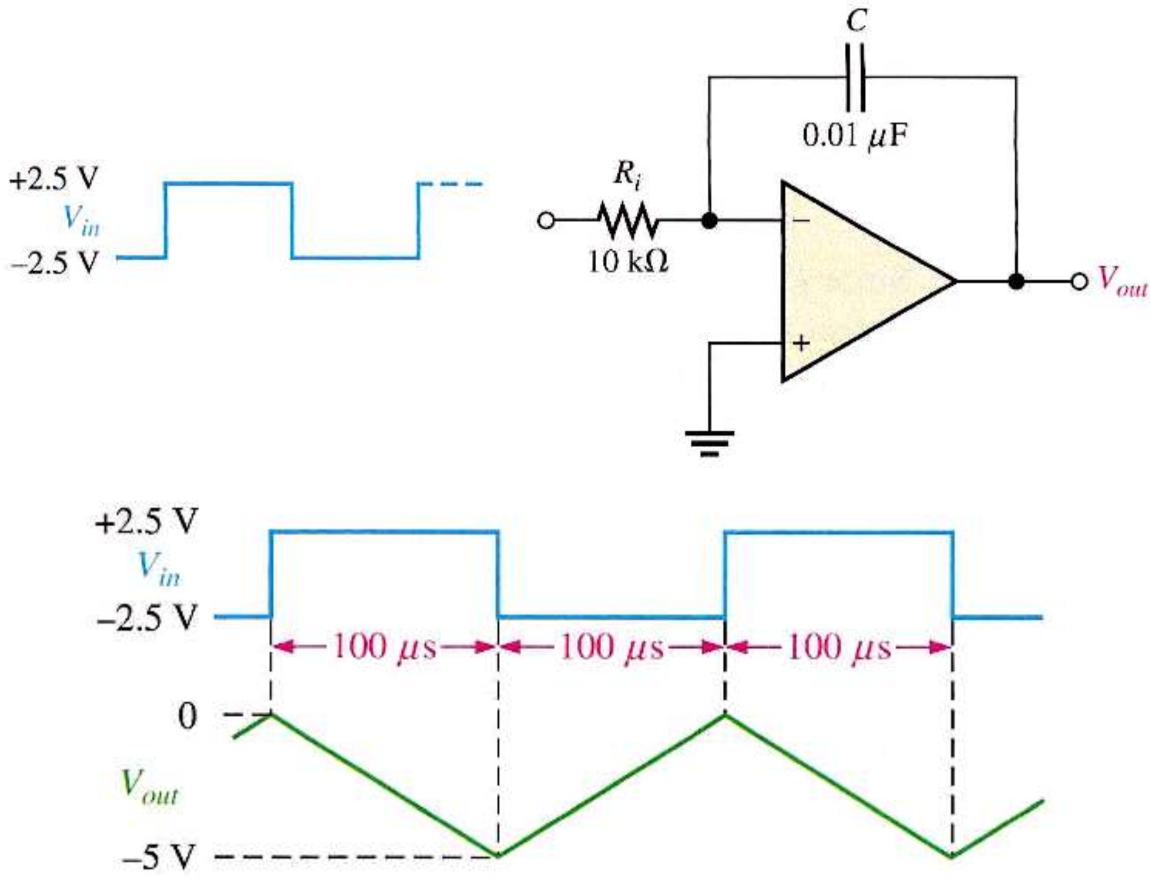
$$i_R = i_C$$

$$\frac{v_i}{R} = -C\frac{dv_o}{dt}$$

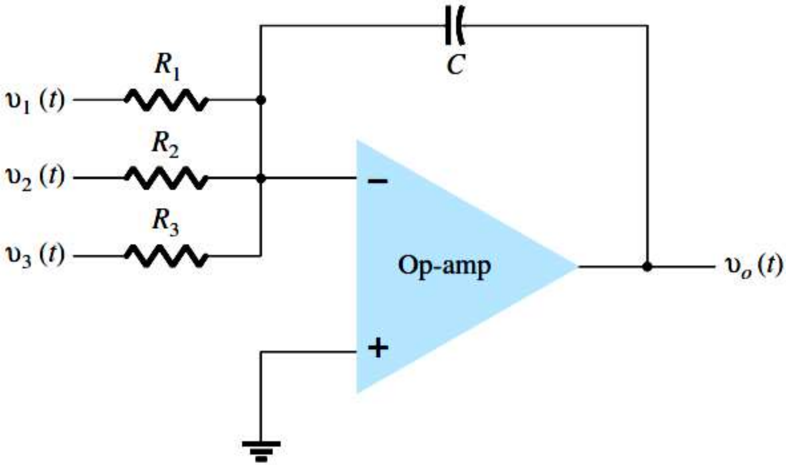
$$dv_o = -\frac{1}{RC} v_i \, dt$$

$$\boxed{v_o = -\frac{1}{RC}\int_0^t v_i(t)\, dt}$$





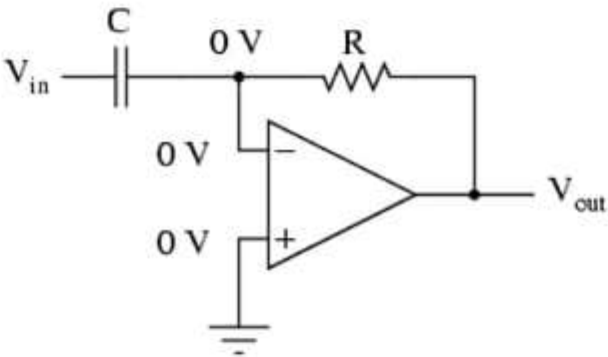
C
0.01 μF
+2.5 V
V_in
−2.5 V
R_i
10 kΩ
V_out
+2.5 V
V_in
−2.5 V
100 μs
100 μs
100 μs
0
V_out
−5 V



$$v_o(t) = -\left[\frac{1}{R_1C}\int v_1(t)\,dt + \frac{1}{R_2C}\int v_2(t)\,dt + \frac{1}{R_3C}\int v_3(t)\,dt\right]$$

Differentiating Circuit

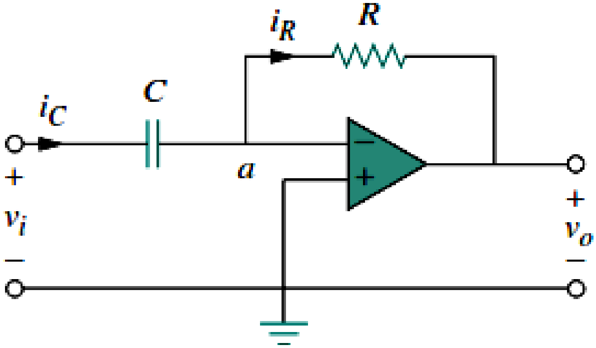
Differentiator



- Input resistor of inverting op-amp is replaced with a capacitor
- Signal processing method which accentuates noise over time
- Output signal is scaled derivative of input signal

$$V_{out} = -RC \frac{dv_{in}}{dt}$$

Differentiator



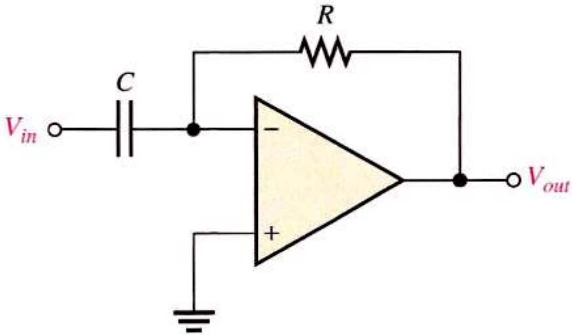
$$i_R = -\frac{v_o}{R}$$

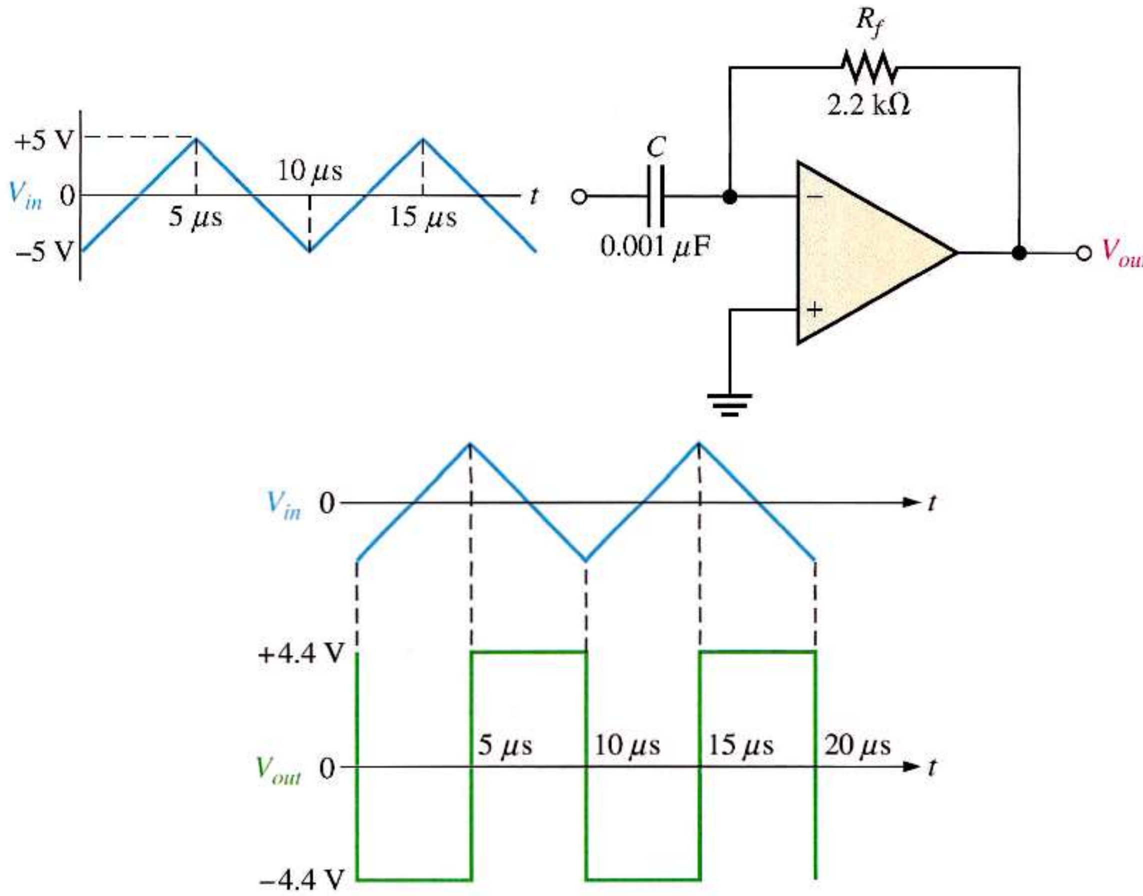
$$i_C = C\frac{dv_i}{dt}$$

$$i_R = i_C$$

$$-\frac{v_o}{R} = C\frac{dv_i}{dt}$$

$$\boxed{v_o = -RC\frac{dv_i}{dt}}$$





+5 V
10 μs
V_{in} 0
t
5 μs
15 μs
−5 V
R_f
2.2 kΩ
C
0.001 μF
V_{out}
V_{in} 0
t
+4.4 V
5 μs
10 μs
15 μs
20 μs
V_{out} 0
t
−4.4 V